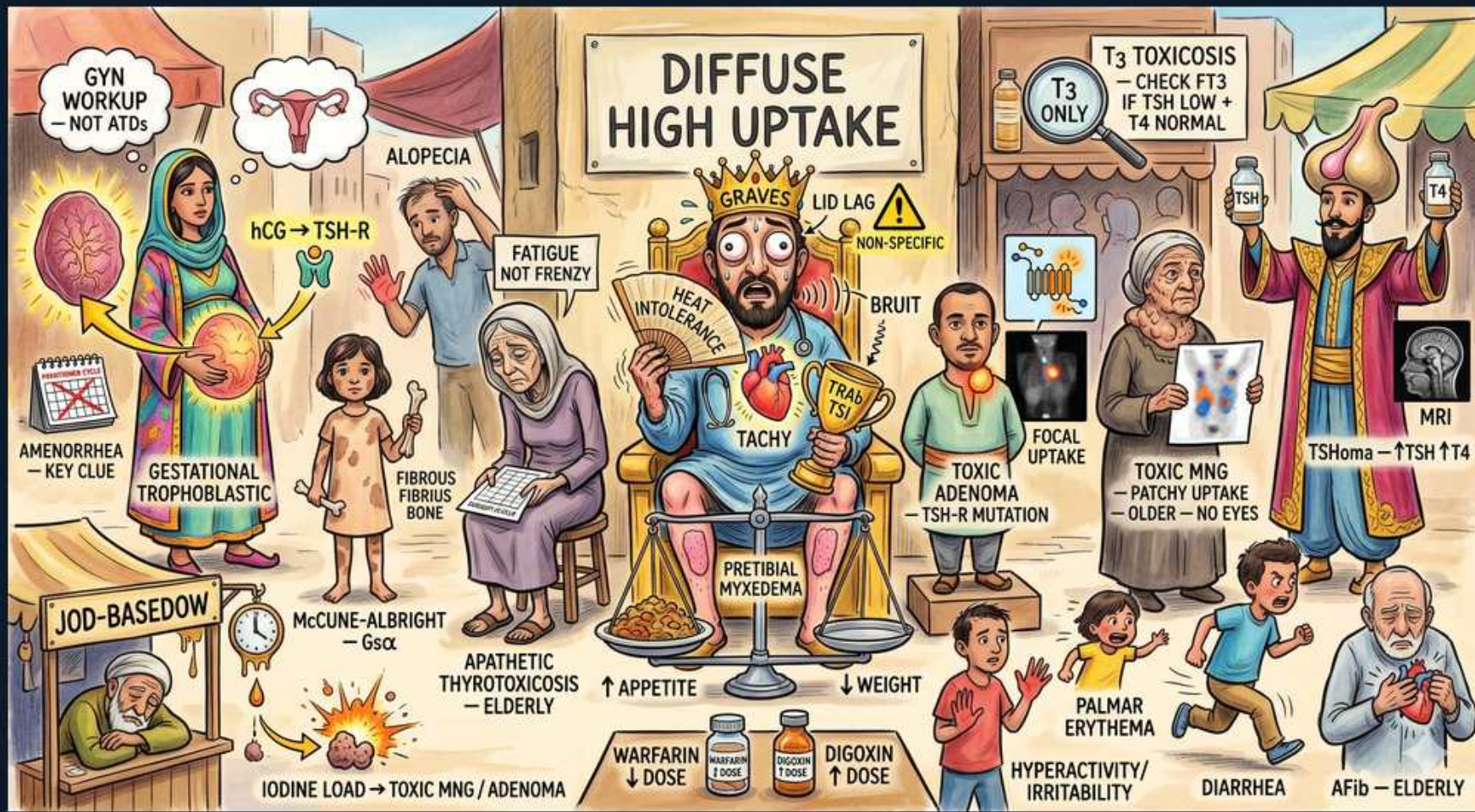


Thyroid — Hyperthyroidism & Graves'



1. Graves' = diffuse high uptake

WHAT YOU SEE

Banner 'DIFFUSE HIGH UPTAKE', Graves king

WHAT IT ENCODES

Graves' disease is the most common cause of hyperthyroidism (esp. women 20–40) and shows DIFFUSE, homogeneously increased radioactive iodine uptake (RAIU) on scan — the whole gland is autonomously stimulated. The uptake scan separates high-uptake causes (Graves', toxic adenoma, toxic MNG) from low-uptake causes (thyroiditis, exogenous/iodine).

BOARD TRAP

Expecting patchy or focal uptake in Graves' — wrong; Graves' is DIFFUSE. Patchy = toxic MNG, single hot focus = toxic adenoma.

2. TRAb antibody

WHAT YOU SEE

'hCG → TSH-R / TRAb' cue

WHAT IT ENCODES

Graves' is driven by stimulating TSH-receptor antibodies (TRAb, also called TSI) that bind and activate the TSH receptor, causing autonomous hormone production and the diffuse uptake. TRAb is the diagnostic and the pathogenic antibody; it crosses the placenta and can cause neonatal thyrotoxicosis. Mechanism mnemonic: TRAb = TSH-Receptor Antibody = 'Turbo' on the receptor.

BOARD TRAP

Naming anti-TPO as the diagnostic antibody for Graves' — wrong; anti-TPO is the Hashimoto's antibody (and is often also positive in Graves' but is not what defines it). TRAb/TSI is the Graves'-specific stimulating antibody.

3. Graves' ophthalmopathy

WHAT YOU SEE

Bug-eyed king, 'LID LAG' arrow (NON-SPECIFIC)

WHAT IT ENCODES

Graves' orbitopathy (exophthalmos/proptosis, diplopia, periorbital edema) is an autoimmune process: TRAb-activated fibroblasts deposit glycosaminoglycans and fat in the orbit — it is SPECIFIC to Graves' and can progress independently of thyroid status. (Lid lag and stare from sympathetic overactivity are non-specific and occur in any thyrotoxicosis.) Smoking worsens it; severe disease is treated with glucocorticoids/orbital radiation.

BOARD TRAP

Attributing true infiltrative ophthalmopathy (proptosis) to a toxic nodule or MNG — wrong; only Graves' causes orbitopathy. Conversely, do not call non-specific lid lag/stare 'Graves' eye disease.'

4. Pretibial myxedema

WHAT YOU SEE

'PRETIBIAL MYXEDEMA' on shins

WHAT IT ENCODES

Pretibial myxedema (thyroid dermopathy) — non-pitting, waxy, indurated plaques over the shins from dermal glycosaminoglycan deposition — is essentially pathognomonic for Graves' and, like orbitopathy and acropachy, is an extrathyroidal autoimmune manifestation. It is uncommon but high-yield because it confirms Graves'.

BOARD TRAP

Confusing 'pretibial myxedema' (a Graves'/HYPERthyroid skin sign) with generalized myxedema of HYPOthyroidism — opposite diseases, similar word.

5. Toxic adenoma

WHAT YOU SEE

'TOXIC ADENOMA — TSH-R MUTATION', FOCAL UPTAKE

WHAT IT ENCODES

A toxic adenoma is a single autonomously functioning nodule, usually from a somatic activating (gain-of-function) mutation in the TSH receptor (or G_{α}). On scan it is a solitary HOT focus with suppressed uptake in the surrounding normal gland. Treatment: RAI or surgery (lobectomy).

BOARD TRAP

Expecting diffuse uptake in a toxic adenoma — wrong; it is focal/single hot nodule. No autoimmune eye/skin disease (TRAb negative).

6. Toxic MNG

WHAT YOU SEE

'TOXIC MNG — PATCHY UPTAKE, OLDER, NO EYES'

WHAT IT ENCODES

Toxic multinodular goiter develops in OLDER patients with a long-standing goiter; multiple autonomous nodules give a PATCHY ('hot and cold') uptake pattern. No autoimmune features (no orbitopathy/dermopathy, TRAb negative). Often unmasked by an iodine load (Jod-Basedow). Treatment: RAI or surgery.

BOARD TRAP

Looking for ophthalmopathy in toxic MNG — wrong; eye disease is Graves'-specific. Patchy uptake + elderly + no eye signs = toxic MNG, not Graves'.

7. T3 toxicosis

WHAT YOU SEE

'T3 TOXICOSIS — CHECK FT3 IF TSH LOW + T4 NORMAL'

WHAT IT ENCODES

T3 toxicosis: TSH is suppressed but free T4 is normal — you must check FREE T3, which is elevated. It occurs early in Graves' or toxic nodular disease, where preferential T3 secretion precedes T4 elevation. Confirms thyrotoxicosis that a T4-only panel would miss.

BOARD TRAP

Excluding hyperthyroidism because free T4 is normal — wrong; with a suppressed TSH and normal T4 you must measure free T3 to catch T3 toxicosis.

8. Gestational / hCG

WHAT YOU SEE

'hCG → TSH-R', gestational trophoblastic, 'GYN WORKUP — NOT ATDs'

WHAT IT ENCODES

hCG shares an alpha-subunit with TSH and weakly cross-stimulates the TSH receptor; very high hCG (hyperemesis gravidarum, multiple/molar pregnancy, choriocarcinoma) causes transient gestational thyrotoxicosis. It is usually self-limited and resolves as hCG falls; molar pregnancy/trophoblastic disease needs gynecologic management.

BOARD TRAP

Treating gestational/trophoblastic thyrotoxicosis with antithyroid drugs — wrong; it is hCG-driven and transient (manage the underlying pregnancy/molar disease, supportive beta-blockers if needed). RAI is absolutely contraindicated in pregnancy.

9. Jod-Basedow

WHAT YOU SEE

'JOD-BASEDOW', IODINE LOAD → TOXIC MNG / ADENOMA

WHAT IT ENCODES

Jod-Basedow phenomenon: an iodine load (CT contrast, amiodarone, iodine supplementation) triggers hyperthyroidism in a gland with pre-existing autonomous nodules (toxic MNG/adenoma) — the autonomous tissue uses the substrate to overproduce hormone. It is the mirror image of the Wolff-Chaikoff effect (iodine load transiently SUPPRESSES a normal gland).

BOARD TRAP

Confusing Jod-Basedow (iodine → HYPERthyroid in autonomous gland) with Wolff-Chaikoff (iodine → transient HYPOthyroid in normal gland) — opposite responses to the same iodine load.

10. McCune-Albright

WHAT YOU SEE

'McCune-Albright — Gsα', fibrous bone

WHAT IT ENCODES

McCune-Albright syndrome: a postzygotic activating GNAS (Gsα) mutation causes constitutive cAMP signaling → autonomous endocrine hyperfunction (hyperthyroidism, precocious puberty, acromegaly), polyostotic fibrous dysplasia, and café-au-lait macules (irregular 'coast of Maine' borders).

BOARD TRAP

Overlooking the classic triad (fibrous dysplasia + café-au-lait + precocious puberty/autonomous endocrinopathy) and missing the Gsα mechanism shared with toxic adenoma.

11. Apathetic thyrotoxicosis

WHAT YOU SEE

'APATHETIC THYROTOXICOSIS — ELDERLY' / 'FATIGUE NOT FRENZY'

WHAT IT ENCODES

In the elderly, hyperthyroidism can present 'apathetically' — fatigue, depression, weight loss, and cardiac signs (atrial fibrillation, heart failure) rather than the classic hyperkinetic agitation/tremor of younger patients. A blunted, withdrawn presentation does not exclude thyrotoxicosis.

BOARD TRAP

Excluding hyperthyroidism because an older patient is lethargic/withdrawn rather than agitated — wrong; check TSH in any elderly patient with new AFib, unexplained weight loss, or heart failure.

12. AFib in elderly

WHAT YOU SEE

'AFib — ELDERLY' heart

WHAT IT ENCODES

New-onset atrial fibrillation should prompt thyroid testing — hyperthyroidism is a reversible, classic cause (excess thyroid hormone increases beta-adrenergic sensitivity and automaticity). Hyperthyroidism also raises the thromboembolic and rate-control challenge of AFib.

BOARD TRAP

Anticoagulating/rate-controlling new AFib without checking TSH — wrong; missing thyrotoxicosis as the reversible driver.

13. ↑ Warfarin & digoxin effect

WHAT YOU SEE

'↓WARFARIN DOSE / ↓DIGOXIN DOSE'

WHAT IT ENCODES

Hyperthyroidism accelerates metabolism of clotting factors (potentiating warfarin → may need a LOWER warfarin dose) and increases digoxin clearance/volume of distribution while raising arrhythmia risk — so doses of warfarin and digoxin must be re-evaluated as thyroid status changes. (When the patient becomes euthyroid, requirements rise again.)

BOARD TRAP

Forgetting that thyroid status alters warfarin and digoxin requirements — failing to recheck INR/levels as the patient is treated toward euthyroidism.

14. Weight loss + ↑appetite

WHAT YOU SEE

'↓WEIGHT, ↑APPETITE'

WHAT IT ENCODES

Hypermetabolism causes weight loss DESPITE an increased appetite — a hallmark that distinguishes thyrotoxicosis from most other causes of weight loss (where appetite falls). Accompanied by heat intolerance, sweating, tremor, palpitations, and hyperdefecation.

BOARD TRAP

Attributing weight loss with preserved/increased appetite to malignancy or diabetes without checking TSH — the appetite-up/weight-down combination is a thyrotoxicosis flag.

15. Thyroid bruit

WHAT YOU SEE

'BRUIT' over the gland

WHAT IT ENCODES

An audible thyroid bruit (and palpable thrill) reflects the markedly increased vascularity of the hyperstimulated Graves' gland — a physical-exam sign specific to Graves' hyperthyroidism, not seen in nodular disease or thyroiditis.

BOARD TRAP

Dismissing a thyroid bruit as a transmitted carotid sound — a true thyroid bruit points specifically to Graves' hypervascularity.

16. Hyperdefecation / diarrhea

WHAT YOU SEE

'DIARRHEA' figure

WHAT IT ENCODES

Increased gut motility from thyroid hormone excess causes hyperdefecation/diarrhea — part of the hypermetabolic syndrome along with heat intolerance and weight loss. (In hypothyroidism the opposite — constipation — occurs.)

BOARD TRAP

Reading 'diarrhea' in a thyrotoxic patient as primary GI disease and missing the hypermetabolic picture.

17. Palmar erythema / warm moist skin

WHAT YOU SEE

'PALMAR ERYTHEMA' hand

WHAT IT ENCODES

Peripheral vasodilation from thyroid hormone excess produces warm, moist, velvety skin with palmar erythema — a peripheral sign of the hyperdynamic, high-output circulatory state of thyrotoxicosis.

BOARD TRAP

Equating palmar erythema only with chronic liver disease — in a hyperthyroid context it reflects the hyperdynamic circulation, not necessarily cirrhosis.
